

Data Visualization

matplotlib

Previous → how to collect, clean and Multiple the data using numpy and pandas

Definition:

It's the graphical representation of data using charts, graph and plots. Instead of reading thousands of numbers, we can understand the data ~~with~~ visually.

We can Comparison

Where we use:

Business Analyst

Bank

weather

gen ai

Matplotlib:
It's a python library used to create
charts, graphs and plots for data
visualization.

Why use Matplotlib:

It helps us:

Understand the data quality

Identify trends

Compare values

detect outliers (isolated)

present visually (reports)

installing matplotlib:

pip install matplotlib

Importing

import matplotlib.pyplot as plt

Matplotlib $\Rightarrow$ library

pyplot $\Rightarrow$ Module used for plotting

plt $\Rightarrow$ Alias

Line graph:

import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5] $\rightarrow$ time, date

y = [10, 20, 30, 40, 50] $\rightarrow$ sales, profit

plt.plot(x, y)

plt.show() $\rightarrow$ Appear the graph

x-axis

time, date, months, categories

y-axis

Sales, profit, marks, temperature

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y-axis

Sales, profit, marks, temperature

Adding title:

```
plt.title("students marks")
```

x-axis label

```
plt.xlabel("subjects")
```

y-axis label

```
plt.ylabel("marks")
```

ex:

```
import matplotlib.pyplot as plt
```

```
subject = ["themarks", "science", "english"]
```

```
marks = [80, 90, 75]
```

```
plt.plot(subject, marks)
```

```
plt.title("subject marks")
```

```
plt.xlabel("subject")
```

```
plt.ylabel("marks")
```

```
plt.show()
```

How to change line color:

```
plt.plot(x, y, color="red")
```

other colors:

blue, green, black, orange, purple.

changing the line style:

```
plt.plot(x, y, linestyle="--")
```

designs:

--
--
--
:

changing the markers:

```
plt.plot(x, y, marker="o")
```

Markers:

o, s, +
*, ^, x

`plt.plot(x, y, color="green", marker="o",
linestyle="--")`

`plt.show()`

Grid operation

* Makes the graph easier to read

`plt.grid()`

Figure Size:

`plt.figure(figsize=(8, 5))`

It changes the graph size

Legend

Helps to compare two students

~~plt.plot~~

`plt.plot(x, marks1, label="John")`

`plt.plot(x, marks2, label="Sara")`

`plt.legend()`

`plt.show()`

Qn:

1. Create a line graph showing marks of five students?

2. Plot monthly temperature?

3. Add title x-axis label

4. Change color, linestyle

5. Display grid

Different types of charts

* Choosing the correct graph is important

* Different graphs are used for different purpose